Crime Rate Prediction Using Machine Learning

Predicting ViolentCrimesPerPop from socio-economic, demographic, and law-enforcement indicators.

CS-UY 4563 | Final Project | Tyler Huynh and Martha McQuillan

Dataset

2,215

communities before cleaning

Feature scale

~120

community-level indicators

Task

Regression

target scaled from 0 to 1

UCI Communities and Crime dataset; target is log1p-transformed during modeling.

The prediction problem

Given a community profile, estimate its violent crime rate.

Why it matters

- Can identify higher-risk communities.
- Can inform public-safety resource allocation.
- Shows links between poverty, inequality, and crime.

What the data contains

- Income, poverty, unemployment, and education.
- Population age groups and demographics.
- Some law-enforcement staffing indicators.

Cleaning made the modeling comparison fair

The same train/validation/test discipline was used across Ridge, neural networks, and KNN.

Split

70 / 15 / 15

train, validation, test

Target

log1p(y)

compresses right skew

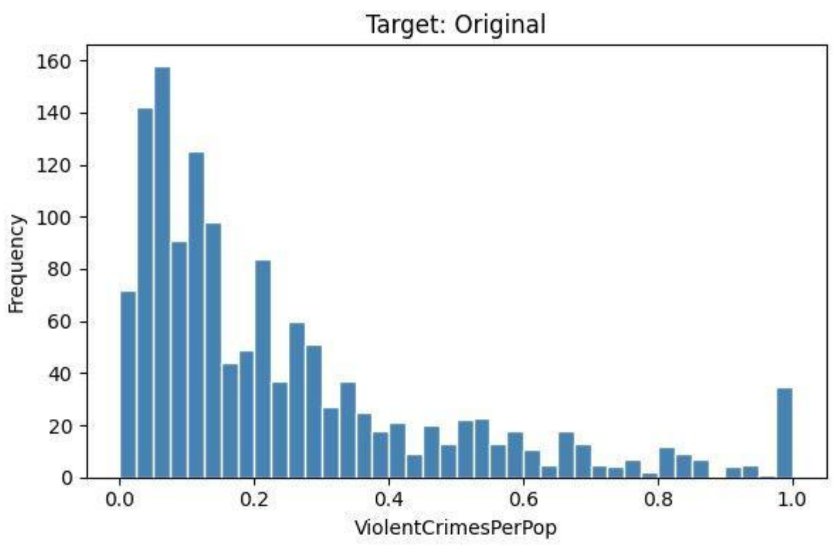
Missing data

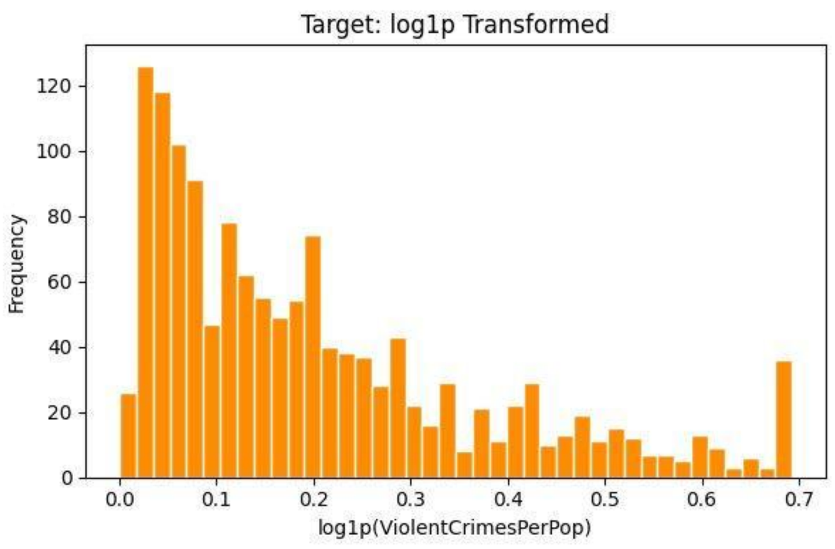
>50%

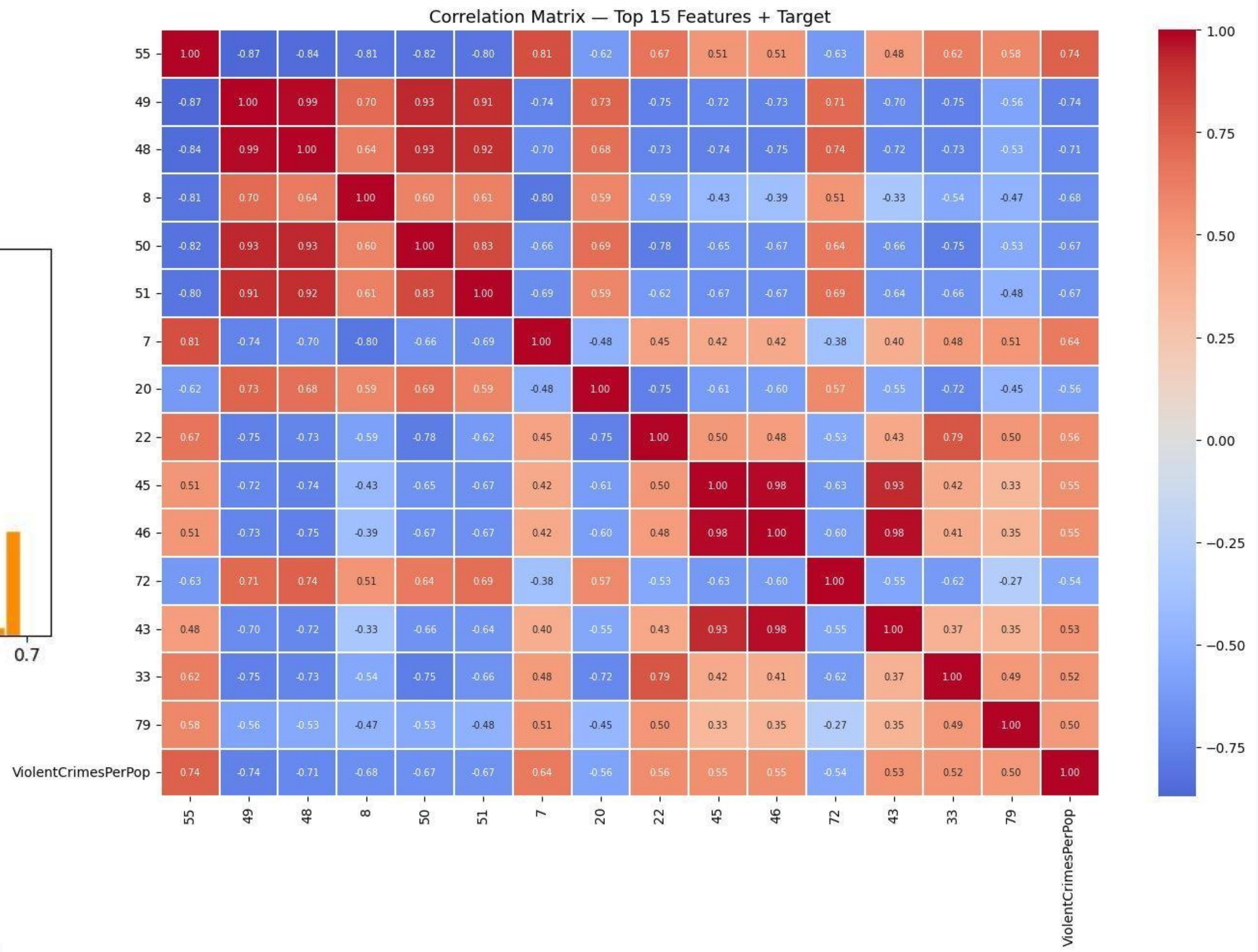
feature columns dropped

EDA shaped the transformations

- The target distribution was heavily right-skewed.
- A small group of features had extremely high missingness.
- Strong feature relationships were curved, motivating polynomial tests.
- Highly correlated predictors motivated PCA.







We compared models and feature spaces

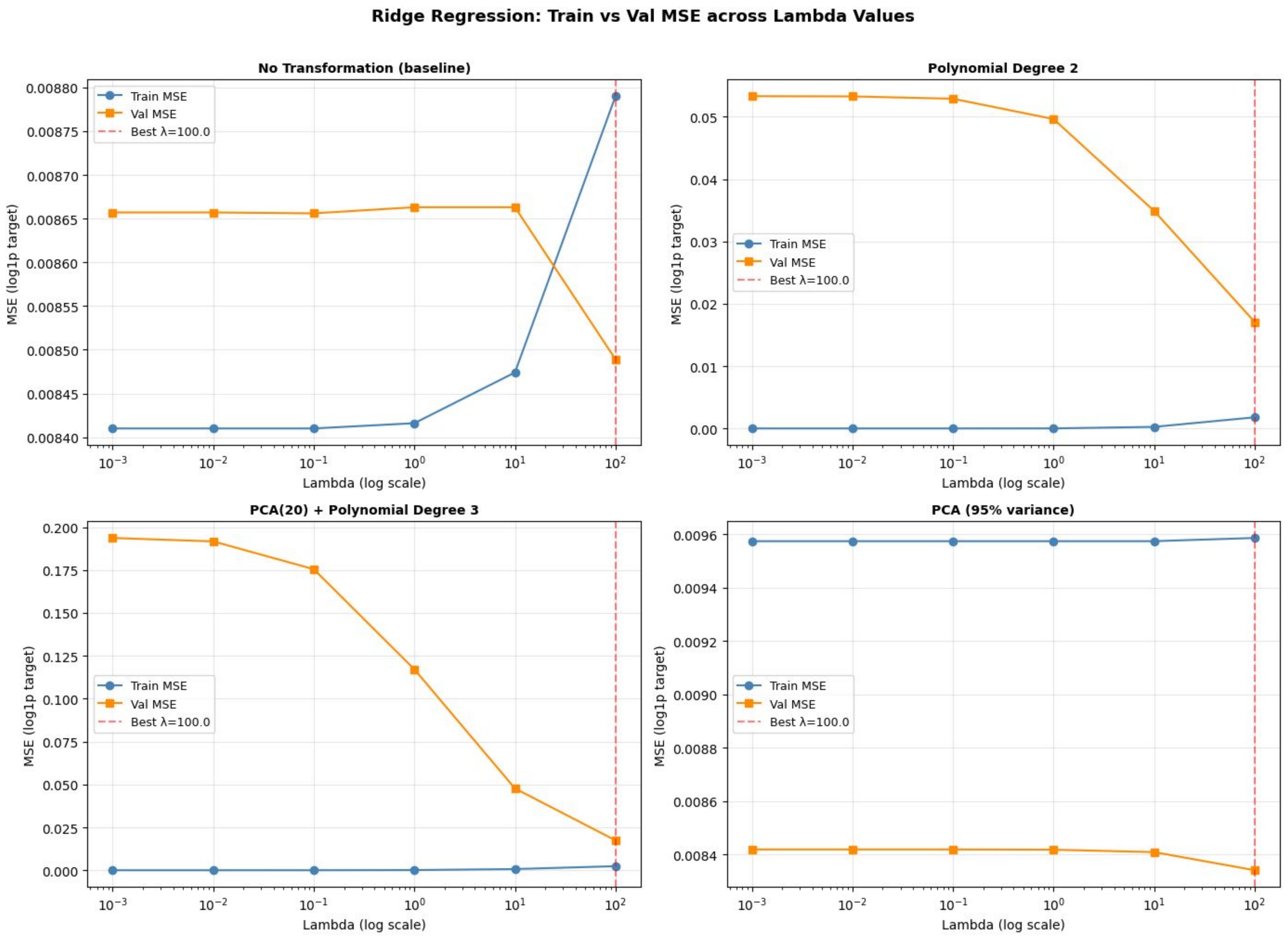
Each model family tested the same basic question: does transforming the feature space improve generalization?

Ridge Regression

24+

lambda: 0.001 to 100

Linear baseline plus transformed feature spaces.



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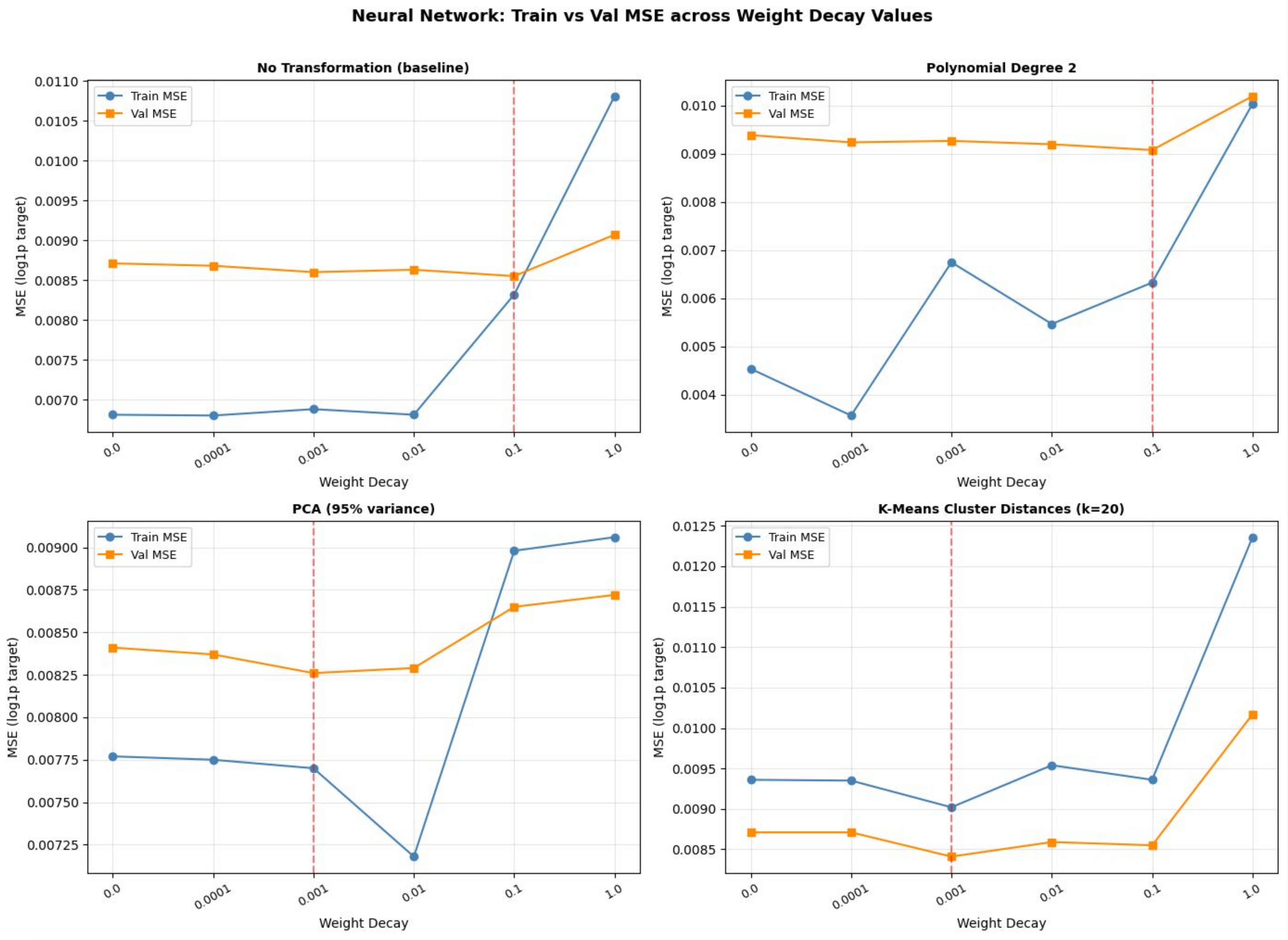
Neural Network

24+

weight decay: 0 to 1.0

Two hidden layers for transformations; early stopping.



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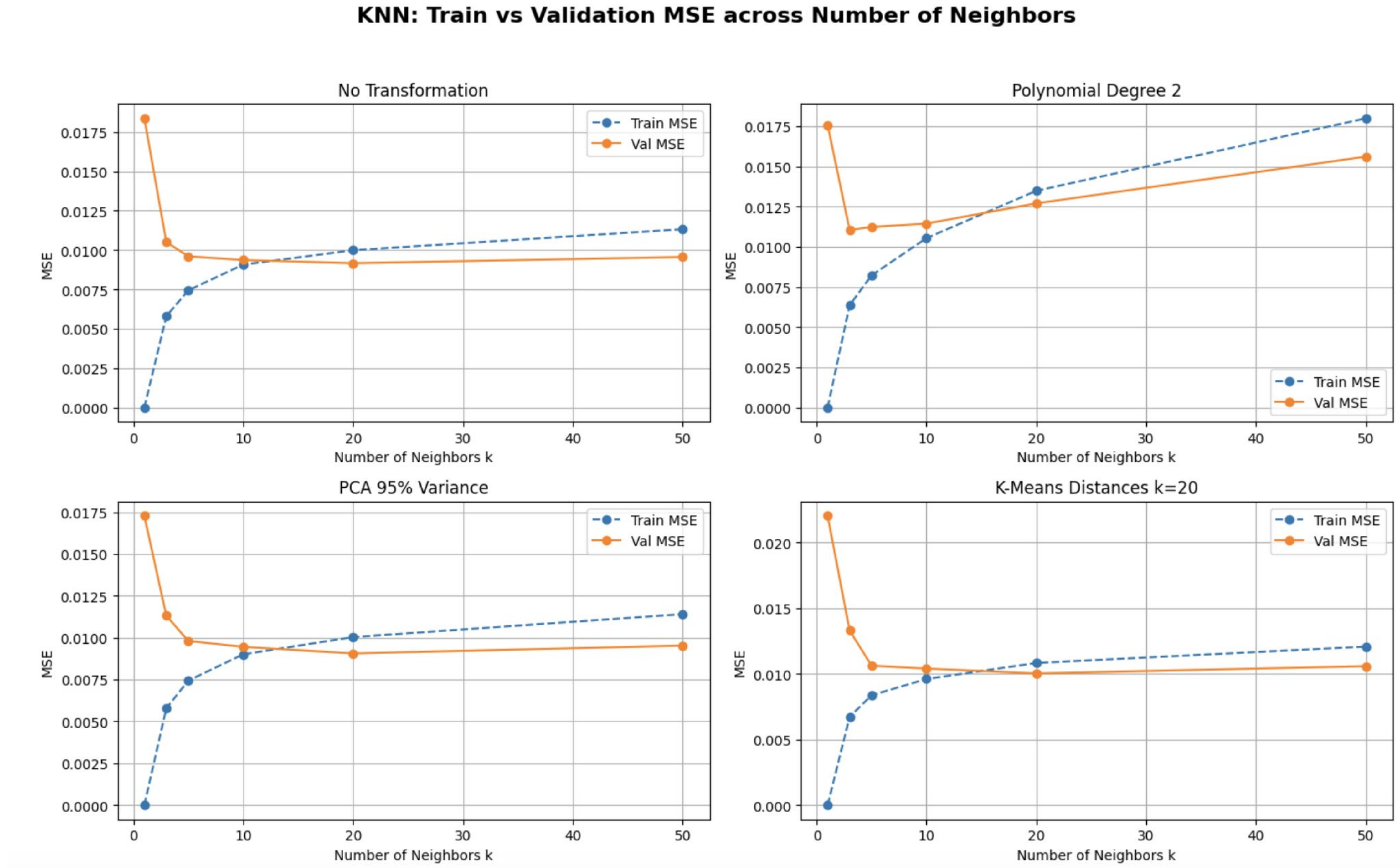


KNN Regression

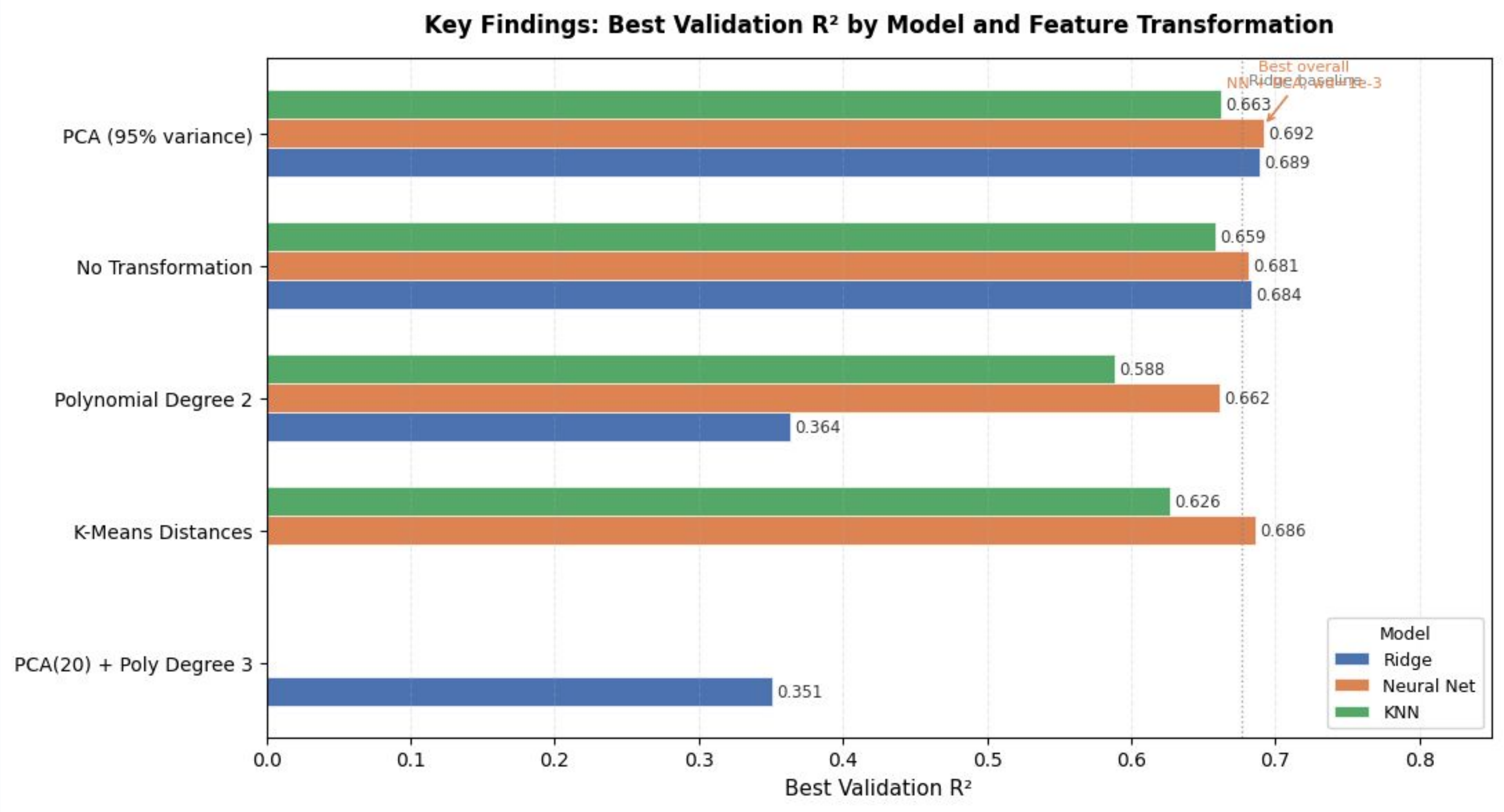
24

neighbors: 1 to 50

Complexity controlled by neighborhood size.

Best validation R^2 by model

The neural network was best overall, but the main story is that PCA helped every model family.



Main takeaway

PCA produced the strongest or most stable validation performance across all three approaches.

Best overall: neural network with PCA and weight decay = 1e-3.

Why PCA was the most reliable transformation

The dataset contained many correlated predictors, which made raw distance and coefficient estimates noisier.

Before PCA

- Many features carried overlapping information.
- Some feature pairs had correlations above 0.90.
- Distance-based models were affected by redundancy.

After PCA

- Correlated features became independent components.
- Noise and dimensionality were reduced.
- Validation results became more stable.

Regularization exposed the bias-variance tradeoff

The transformations behaved very differently as model complexity changed.

Polynomial features

Captured possible curves but added too much complexity. Ridge and KNN overfit badly, especially at low regularization.

PCA features

Reduced redundancy instead of adding terms, so performance was strong and less sensitive to hyperparameters.

K-Means distances

Encoded community clusters well for the neural network, but did not preserve enough local detail for KNN.

Conclusion

The best model was not the most complex one; it was the model with the cleanest representation.

What we found

- Socio-economic structure is meaningfully predictive of crime rate.
- PCA was the strongest and most consistent transformation.
- Polynomial expansion added more noise than useful signal.
- The best validation R^2 was 0.692 from NN + PCA.

Next steps

- Report error in original crime-rate units.
- Evaluate the held-out test set once.
- Try an ensemble of Ridge, NN, and KNN predictions.